



Augmentation



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Why Augmentation is implemented

- Documents that are relevant to the user hardly ever contain exactly the same query terms
 - synonyms (*car* and *automobile*)
 - syntactically different (*USA*, *U.S.A.* and *United States of America*)
- Users always have a limited ability to express their information needs
- Natural language is inherently ambiguous

These are the main reasons why query result does not satisfy the user expectation

Solution: **Augmentation**

What is augmentation?

Augmentation is the process of enhancing the relevance and **accuracy** of search results by providing additional context or data beyond the initial user query.

How Augmentation is implemented

- By using Lexical-Semantic Relations (discussed in the follows)
- By using Relevance Feedback (discussed in the follows)
- By using Word Embedding (discussed in the follows)
- By using Statistical Relations (Latent Semantic Indexing, not discussed in this course)

Lexical Semantic Relations

Lexical-Semantic Relations

This method expands a term with a list of terms that somehow are **semantically** related to the original one. Some relations can be:

- Synonyms: same or very similar meanings (i.e. *car* and *automobile*)
- Antonyms: opposite meanings (i.e. *hot* and *cold* are antonyms)
- Hyponyms/Hypernyms: “is-a” relation (i.e. *dog* is an hyponym of *mammal*, *mammal* is an *hypernym* of *dog*)
- Holonyms/Meronyms: “part-whole” relation (i.e. *engine* is a meronym of *car*, and *car* is a holonym of *engine*)
- Pertainyms: (i.e. *Scientific* is related to *science* through pertainym)

Lexical-Semantic Relations: WordNet

WordNet (<https://wordnet.princeton.edu/>) is a large lexical database for english terms which describes relations among terms in approximately 10 hierarchies.

The last version of WordNet (version 3) consists of **three databases**, one each for **nouns** and **verbs** and a third for **adjectives and adverbs**

They contain:

- 117,798 nouns
- 11,529 verbs
- 22,479 adjective and 4,481 adverbs

Lexical-Semantic Relations: WordNet

Given a term t , WordNet provides a list of terms that are related with t : $\{t_1, t_2, \dots\}$

Such list is named **synset**

All the terms of the synset or a subset of them can be used to augment the original query.

Lexical-Semantic Relations: Global WordNet

There is a WordNet database for almost all languages:

<https://globalwordnet.org/>

Global WordNet is an active project that every year organizes a conference around the World. The 13th Global Wordnet Conference was held last January 2025 in Pavia.

Lexical-Semantic Relations: Italian WordNet

WordNet for italian language

<https://www.ilc.cnr.it/progetti/italwordnet-2/>

Relevance Feedback

Relevance Feedback

The idea behind the Relevance Feedback is that the user provides his feedback to improve the final result:

1. The user issues a (short, simple) query
2. The system returns an initial set of retrieval results
3. The user marks some returned documents as relevant or non-relevant
4. The system computes a better representation of the information need based on the user feedback
5. The system displays a revised set of retrieval results

Relevance Feedback

Step number 4 seems to be tricky: how can the system get a better representation of user's need?

The idea is to move the vector representing the user query toward the centroid representing the set of relevant documents and far away from those representing the non relevant documents.

This algorithm is known as **Rocchio's Algorithm**

Relevance Feedback: Rocchio's Algorithm

Let $\vec{q}_0$ be the original user query, represented by a vector

Let D_r be the set of documents labeled as relevant by the user

Let D_{nr} be the set of documents labeled as non relevant by the user

Let $\vec{d}_j$ be a document, represented by a vector

Let α, β and γ three weights

$$\vec{q}_m = \alpha \vec{q}_0 + \beta \frac{1}{|D_r|} \sum_{\vec{d}_j \in D_r} \vec{d}_j - \gamma \frac{1}{|D_{nr}|} \sum_{\vec{d}_j \in D_{nr}} \vec{d}_j$$

Relevance Feedback: Rocchio's Algorithm

Let $\vec{q}_0$ be the original user query, represented by a vector

Let D_r be the set of documents labeled as relevant by the user

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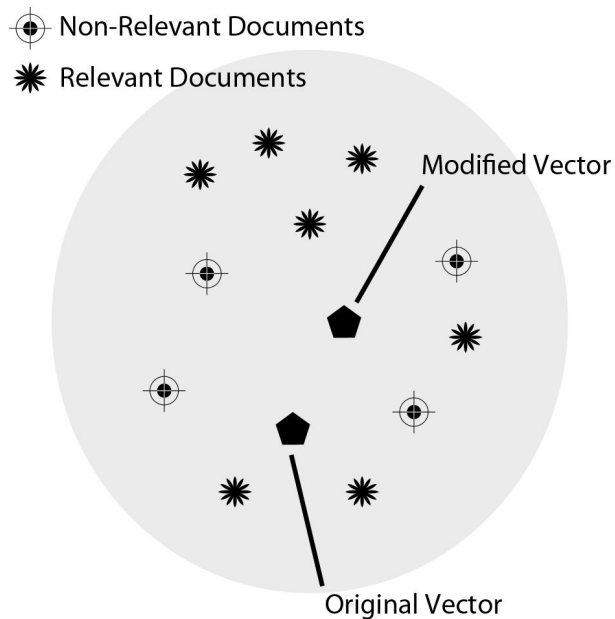
Let $\vec{d}_j$ be a document, represented by a vector

Let α , β and γ three weights

$$\vec{q}_m = \alpha \vec{q}_0 + \beta \left[\frac{1}{|D_r|} \sum_{\vec{d}_j \in D_r} \vec{d}_j \right] - \gamma \left[\frac{1}{|D_{nr}|} \sum_{\vec{d}_j \in D_{nr}} \vec{d}_j \right]$$

Centroid of D_r Centroid of D_{nr}

Relevance Feedback: Rocchio's Algorithm



Relevance Feedback: Rocchio's Algorithm

Experimental results shows that:

- Relevance feedback can improve both recall and precision
- Positive feedback are more valuable than negative feedback
 - Reasonable values are:
 - $\alpha = 1$
 - $\beta = 0.75$
 - $\gamma = 0.15$

Word Embedding Expansion

Word Embedding Expansion

Basic Idea:

- Load pre-trained embeddings using Word2Vec or FastText
- For each term in the original query:
 - Find top-k similar words via cosine similarity
- Filter out irrelevant candidates (e.g., using part-of-speech tags, frequency thresholds)
- Add selected words to the query (possibly weighted lower than original terms)

Implementing Augmentation

When Augmentation is implemented

- **At query time (query expansion):** during query time, a term is **expanded** with multiple terms belonging to the equivalence class. The query term is then a disjunction of several postings list (*car OR automobile*)
- **At indexing time (document expansion):** if the document contains *car*, we index the document even under *automobile*

Pros and Cons

- **At query time (query expansion):** this method requires a query expansion dictionary and requires more processing at query time
- **At indexing time (document expansion):** this method requires more space for storing postings.